

RESEARCH METHODOLOGY

WEEK 4: TECHNICAL PRESENTATION AND TECHNICAL WRITING

Last-Minute MCQ and Marks-Grabbing Notes

Purpose: A compact revision document built from the Week 4 notes you provided, the current Week 4 assignment questions visible in your screenshots, and older NPTEL Week 4 assignment material.

Exam focus: definitions, distinctions, application-based MCQs, True/False traps, multi-select questions, and wording-based traps.

Font: Times-style formal layout; title 16 pt, section headings 14 pt, body 12 pt.

The official NPTEL course page identifies Week 4 as “Technical presentation and writing,” with the lessons “How to make Technical presentation” and “Technical Writing.”

Source: Official NPTEL course page: <https://nptel.ac.in/courses/121106007>

PART I — TECHNICAL PRESENTATION

Core idea: A technical presentation is a live, interactive, audience-focused communication of technical information. It should be adapted to the audience and the available time rather than being a paper read aloud.

1. Presentation versus Journal Paper

Technical Presentation	Journal Paper
Live and interactive	Static and formal record
Dynamic and flexible	Structured for independent reading
Audience-focused	Completeness-focused
Can be adapted for different audiences	Becomes a permanent part of the literature

Memory: Presentation = live + flexible + audience-focused. Paper = static + formal + complete.

2. Two Primary Constraints

- **Audience:** determines the depth and style of the presentation.
- **Duration:** determines the scope and pacing.
- A professional presenter respects both constraints.
- If the available time is reduced, do not merely speak faster; select the most critical content.

3. Pacing and Agility

- Rule of thumb in the notes: approximately one slide per minute.
- Fewer slides can be better when they allow enough time for explanation.
- Agility means being able to adjust the talk when the available time changes.
- A 40-minute talk cut to 20 minutes should be shortened by selecting the essential slides, not by racing through all slides.

4. Connecting with the Audience

- Use eye contact.
- Use conversational delivery.
- Design slides that aid comprehension.
- Tailor the message to the audience's background and motivation.
- The goal is not merely to show data; it is to make the audience understand it.

5. Slide Design: Less Is More

- Prefer one main idea per slide.
- Break complicated information across several slides.
- Use whitespace to direct attention.
- Avoid overcrowded slides containing too many graphs, photographs, experimental details, and paragraphs.

6. Reading Slides, Fonts and Effects

- Do not write full paragraphs and read them aloud.
- Use bullet points as cues or signposts.

- Slides should support the speaker, not replace the speaker.
- Avoid distracting animations and sound effects.
- Use special effects judiciously.
- Use about 1–2 standard, readable fonts consistently.

7. Choosing the Correct Illustration

Illustration	Best use
Table	Specific data points / exact values
Graph	Trends, comparisons, relationships
Photograph	Realism and visual context
Drawing / Schematic	Simplifying complex systems and showing components

8. Graphs and Tables: What Must Be Present?

- Graph: title, clearly labelled X and Y axes, units, and a legend when multiple data sets are shown.
- Table: clear headings, units, consistent formatting, and meaningful significant digits.
- Do not report calculator digits blindly; report meaningful precision.
- Use a graph when the important message is a trend, comparison, or relationship.
- Use a table when exact values are the important information.

9. Photographs and Scale

- Avoid distracting backgrounds and unrelated equipment.
- Use an overall photograph for context and a close-up for important details when useful.
- A technical image should provide appropriate scale.
- A micrograph can use a scale bar such as “20 nm”; a familiar object such as a coin can provide scale in a photograph.

10. Magnification versus Resolution

Magnification: makes an object or image appear larger.

Resolution: refers to the ability to distinguish fine details.

Key MCQ: High magnification does not necessarily imply high resolution.

11. Beginning and Ending

- Start with an outline or roadmap.
- Revisit the roadmap between major sections when useful.
- End with a strong summary of the most important points.

PART II — TECHNICAL WRITING

Core idea: Technical writing is formal scientific communication designed for efficient, clear transmission of research findings. The objective is not suspense; it is efficient understanding.

1. Technical Writing

Technical writing is a formal style used by researchers to communicate new findings, primarily through academic articles. It provides a mechanism for formal credit, contribution to scientific knowledge, and influence on policy or industry.

2. Peer Review

- A submitted article is evaluated by independent experts in the same field.
- The experts are described in the notes as anonymous.
- Peer review checks the novelty, significance, and quality of the research.
- Do not confuse peer review with simple proofreading.

3. Efficiency — The Central Principle

The notes describe efficiency as the single most important goal of technical writing. Every element should help the reader understand the work as quickly and clearly as possible.

- Use simple, direct language.
- Avoid unnecessary repetition.
- Avoid ambiguity.
- Avoid unnecessarily long or complicated sentences.
- Do not use language merely to impress the reader.

4. The Structure of a Scientific Paper

Section	Main question / purpose
Title	What is the research area and what is the specific contribution?
Abstract	What is the problem, method, key result, and conclusion?
Introduction	What is the background, why is it important, and what is novel?
Methods	How was the work done, with enough detail for reproduction?
Results	What was observed?
Discussion	Why did it happen and what does it mean?
Conclusion	What are the most important findings?
Future perspective	What are logical next steps for research?

5. Title

- The title should be precise.
- It should identify the broad research area.
- It should also indicate what is unique or novel about the work.
- A vague title may fail to identify either the research area or the specific contribution.

6. Abstract

- Concise and standalone summary.

- Contains the major findings and conclusions.
- Includes the problem, methods, key results, and main conclusion.
- It is designed for a reader who may not read the entire paper.
- Do not hide the key result merely to create suspense.

7. Introduction

- Provides necessary background and context.
- Explains what the work is.
- Explains why it is important.
- Establishes novelty by comparing the work with previous research and identifying what has not yet been explored.
- Introduces specific models, theories, or background knowledge needed by the reader.

8. Methods / Experimental Details

- Think of Methods as a recipe.
- Provide enough detail for another researcher to reproduce the work.
- Nothing essential should be hidden.

9. Results versus Discussion

Results	Discussion
WHAT happened?	WHY did it happen? / SO WHAT?
Factual and objective	Interpretation and analysis
States observations	Explains implications and builds the argument

10. Conclusion and Future Perspective

- Conclusion summarizes the most important findings.
- Unlike the abstract, the conclusion is aimed at readers who have read the paper.
- The conclusion may suggest logical next steps for future research.
- Keep the distinction clear: Abstract = standalone overview; Conclusion = final synthesis for an informed reader.

11. Language Traps

- Avoid ambiguous words such as “this” when the referent is unclear.
- Avoid vague or arrogant phrases such as “It is obvious that...”
- Sentence length should be controlled so that the reader does not lose the beginning before reaching the end.
- Scientific documents should communicate efficiently and clearly.

PART III — CURRENT WEEK 4 ASSIGNMENT QUESTIONS SHARED BY YOU

The following questions are transcribed from the three screenshots you provided. The accepted answers shown in the screenshots are reproduced, with concise explanations for last-minute revision.

1. The title of a paper is “Battery design and its impact”. This title:

- A) Identifies the area of research clearly
- B) Indicates what is unique about the research
- C) Does not identify the area of the research clearly
- D) Does not indicate what is unique about the research

Answer: C and D

Why: The accepted answer in your screenshot marks both weaknesses. The title is too broad to clearly identify the research area and does not communicate the specific novel contribution.

2. In images:

- A) High magnification implies high resolution
- B) High magnification does not necessarily imply high resolution
- C) Low magnification implies high resolution
- D) High resolution implies low magnification

Answer: B

Why: Magnification and resolution are different concepts. Increasing apparent size does not automatically increase the ability to distinguish fine detail.

3. Tables are preferred over graphs:

- A) Always
- B) Never
- C) Only when exact values need to be indicated
- D) Only when trends in data need to be indicated

Answer: C

Why: The accepted answer in your screenshot is C. Tables are useful when the exact numerical values themselves are important; graphs are better for trends, comparisons, and relationships.

4. The length of a document:

- A) Indicates how much you know
- B) Indicates how little you know
- C) Does not indicate anything about your knowledge
- D) Does not indicate anything about the reader's knowledge

Answer: C and D

Why: The accepted answer marks both C and D. Document length is not a reliable measure of either the author's knowledge or the reader's knowledge.

5. The title of a paper should indicate the area of work as well as what is unique about the work.

- A) True
- B) False

Answer: True

Why: This directly follows the title principle: identify the research area and the specific or novel contribution.

6. In scientific documents sentences should be relatively short and to the point.

- A) True
- B) False

Answer: True

Why: Short, direct sentences improve efficiency and reduce the chance that readers lose track of the beginning of a sentence.

7. In an article, the image of a car engine must:

- A) Show the level of detail required for the discussion in the article
- B) Have a marker bar indicating scale
- C) Be in colour and not in black and white
- D) Be in black and white and not in colour

Answer: A and B

Why: The accepted answer in your screenshot marks A and B. The image should show the appropriate detail and indicate scale; colour is not an absolute requirement.

8. Strategies one can use in the middle of a document include:

- A) Chronological strategy
- B) Spatial strategy
- C) Following a variable
- D) Suspense strategy

Answer: A, B and C

Why: The accepted answer marks chronological strategy, spatial strategy, and following a variable. Suspense strategy is not the accepted scientific-writing strategy here.

9. While working with electricity only the voltage poses a danger to the users.

- A) True
- B) False

Answer: False

Why: The statement is too restrictive. The accepted answer in your screenshot is False.

10. “Pouring rain results in clogged drains, and slippery roads. These clogged drains fill further water on to the roads.” Use of the word ‘these’ in the previous sentence is ambiguous.

- A) True
- B) False

Answer: False

Why: The accepted answer in your screenshot is False. Here “these” can be understood as referring to the immediately preceding clogged drains; therefore the statement is not considered ambiguous in the assignment.

PART IV — OLDER NPTEL WEEK 4 QUESTIONS AND ANSWERS

These are older Week 4 questions found in publicly available copies of NPTEL assignment material. They are useful because they reveal the style of traps used in this topic. They should be treated as previous-question practice, not as a guarantee of repetition.

1. **“Impact of mobility on velocity” identifies the area of research and what is unique about it.**

Answer: False

Why: The title is not sufficiently precise about the research area and the unique contribution.

2. **“Low tyre pressure leads to a softer ride and greater wear and tear of the tyre. This is desirable from the perspective of the tyre company.” Use of “this” is ambiguous.**

Answer: True

Why: The referent of “this” can be unclear; technical writing should avoid such ambiguity.

3. **In images, high magnification leads to high resolution.**

Answer: False

Why: High magnification does not necessarily imply high resolution.

4. **Abstract is aimed at an audience that has not yet read the paper.**

Answer: True

Why: The abstract is a standalone overview for a reader who may not read the full paper.

5. **Data in tables must always indicate as many digits as obtained through the calculator.**

Answer: False

Why: Use meaningful and appropriate precision rather than reporting every calculator digit.

6. **A reasonable pace for presentation is 5 slides a minute.**

Answer: False

Why: The presentation must allow the audience enough time to absorb the material; the notes use roughly one slide per minute as a rule of thumb.

7. **In a journal paper, an image of a satellite must indicate the solar panels to confirm that it represents a satellite in orbit.**

Answer: False

Why: An illustration does not need to show every possible component merely to establish its intended context.

8. **The length of sentences should be such that by the time readers reach the end, they should not have forgotten the beginning.**

Answer: True

Why: Sentence length should support clarity and comprehension.

9. **Once you know the result, it is acceptable to state it using “It is obvious that...”.**

Answer: False

Why: The course emphasizes clear, direct language and discourages vague or arrogant phrases such as “it is obvious that...”.

10. **One must give away the best result of the work right at the beginning of the document itself.**

Answer: True

Why: Technical writing prioritizes efficient communication rather than suspense; important information should be communicated early.

Public source used to cross-check these older questions: publicly available copies of NPTEL Week 4 assignment material, including the 2023 assignment.

Reference: <https://www.scribd.com/document/678824483/Research-Methodology-Unit-6-Week-4-Technical-presentation-and-writing>

Additional public Week 4 material: <https://www.scribd.com/document/842946772/W4>

PART V — HIGH-YIELD PRACTICE MCQs

These are additional practice questions created from the supplied Week 4 notes and the assignment patterns. They are practice questions, not claimed as previous-year questions.

1. Which pair correctly matches the section with its main role?

- A) Results — why the result occurred
- B) Discussion — raw observations
- C) Methods — reproducibility
- D) Title — detailed methodology

Answer: C

Why: Methods must contain enough detail for another researcher to reproduce the work.

2. Which illustration is most appropriate when exact numerical values are important?

- A) Table
- B) Graph
- C) Photograph
- D) Schematic

Answer: A

Why: Tables are preferred when exact values need to be indicated.

3. Which statement about an abstract is correct?

- A) It should hide the key result
- B) It is a standalone summary
- C) It is written only for peer reviewers
- D) It contains only background

Answer: B

Why: The abstract is a concise, standalone summary including major findings and conclusions.

4. Which phrase best represents the Results section?

- A) What happened?
- B) Why did it happen?
- C) What should happen next?
- D) Why is the paper novel?

Answer: A

Why: Results present factual observations; interpretation belongs mainly in Discussion.

5. Which phrase best represents the Discussion section?

- A) What happened?
- B) Why and what does it mean?
- C) What is the title?
- D) How long is the paper?

Answer: B

Why: Discussion analyzes results, explains implications, and develops the scientific argument.

6. Which is NOT a recommended presentation practice?

- A) Eye contact
- B) Conversational delivery
- C) Reading full paragraphs from slides
- D) One main idea per slide

Answer: C

Why: Slides should support the speaker, not replace the speaker.

7. A presentation time is unexpectedly cut in half. The best response is to:

- A) Speak twice as fast
- B) Use every slide anyway
- C) Select the most critical content
- D) Remove the conclusion only

Answer: C

Why: Agility means adapting the talk to the available time.

8. Which is most important when using multiple data sets in one graph?

- A) A legend
- B) A decorative background
- C) Many fonts
- D) Sound effects

Answer: A

Why: A legend helps distinguish multiple data series.

9. Which statement about peer review is correct?

- A) It is mainly proofreading
- B) It is evaluation by independent experts
- C) It is performed by the general public
- D) It guarantees the research is correct forever

Answer: B

Why: Peer review evaluates research quality, novelty, and significance through experts in the field.

10. Which statement best describes efficiency in technical writing?

- A) Use difficult vocabulary
- B) Maximize document length
- C) Help the reader understand quickly and clearly
- D) Create suspense

Answer: C

Why: Efficiency is the central communication goal described in the notes.

PART VI — LAST-MINUTE TRAP SHEET

If you see...	Think...
Table	Exact values / specific data
Graph	Trend / comparison / relationship
Results	WHAT happened
Discussion	WHY / SO WHAT
Methods	Reproduce the work
Abstract	Standalone overview; give major findings
Introduction	Background + importance + novelty
Conclusion	Main findings for a reader who read the paper
Title	Research area + unique contribution
Peer review	Independent experts in the field
Efficiency	Fast + clear understanding
Scale bar	Size/context
Magnification	Apparent size
Resolution	Ability to distinguish detail
“It is obvious...”	Avoid in technical writing
Full paragraphs on slides	Avoid; use cues/signposts
5 slides/minute	Too fast; false as a reasonable pace
“Always” / “must”	Read carefully; many traps are overgeneralizations

PART VII — 2-MINUTE REVISION

Technical Presentation: Audience + Duration; one main idea per slide; do not read slides; use appropriate illustrations; label graphs; use meaningful table formatting; provide scale; magnification is not resolution; start with a roadmap; finish with a summary.

Technical Writing: Efficiency; precise title; standalone abstract; introduction establishes novelty; methods enable reproduction; results state WHAT; discussion explains WHY/SO WHAT; conclusion summarizes findings; future perspective suggests next steps; avoid ambiguity and phrases such as “it is obvious that...”.

Core sequence: TITLE → ABSTRACT → INTRODUCTION → METHODS → RESULTS → DISCUSSION → CONCLUSION → FUTURE.

Golden distinction: Results = WHAT. Discussion = WHY/SO WHAT.

Golden distinction for illustrations: Table = exact values. Graph = trends/relationships. Photograph = realism. Schematic = system/components.

SOURCE NOTE

This document is based primarily on the two Week 4 summary PDFs supplied in the conversation and the three screenshots of the current Week 4 assignment supplied by the user. Older-question material was cross-checked against publicly available NPTEL Week 4 assignment copies.

Official course: <https://nptel.ac.in/courses/121106007>

Public older Week 4 assignment copy: <https://www.scribd.com/document/678824483/Research-Methodology-Unit-6-Week-4-Technical-presentation-and-writing>

Public Week 4 material: <https://www.scribd.com/document/842946772/W4>